Plastic Films

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
LDPE Film (General Packaging)	IS 2508:2024 (Polyethylene Films and Sheets — Specification)	ISO 527-3 (Tensile properties for films and sheeting); ISO 4593 (Thickness determination by mechanical scanning); ISO 4591 (Plastic Film & Sheeting – Average Thickness by Gravimetric Method);ISO 7765-1:Free-falling dart impact test for films; ISO 8295:Friction coefficient evaluation;	ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);ASTM D1003:Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics;Thickness (D374/D6988)	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 4593 (Thickness Measurement of Plastic Film and Sheeting); DIN EN ISO 7765-1 (Impact Resistance of Plastic Film by Free-Falling Dart Method)	EN 13432 (Packaging Recoverable Through Composting, where compostable claims apply); JIS Z 1707 (Plastic Films for Packaging); GB/T 10004 (Plastic Composite Films for Packaging)
LDPE Film (Food Packaging)	IS 15609:2005:Polyethylene Flexible Pouches for the Packing of Natural Mineral Water and Packaged Drinking Water; IS 9845:1998:Determination of Overall Migration of Plastic Materials and Articles Intended to Come into Contact with Foodstuffs — Method of Analysis;; IS 2508:2024 (Polyethylene Films and Sheets — Specification);IS 10146:1982(Polyethylene Materials for safe use with Food and Pharma)	ISO 527-3 (Tensile properties for films and sheeting); ISO 4593 (Thickness determination by mechanical scanning); ISO 4591 (Plastic Film & Sheeting – Average Thickness by Gravimetric Method);ISO 7765-1:Free-falling dart impact test for films; ISO 8295:Friction coefficient evaluation;ISO 22000 (Food Packaging / Food Safety Management); ISO 18602 (Packaging and the Environment – Optimization of Packaging System);	ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D6868 (Biodegradable/Compostable Packaging Applications, where applicable); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method);Thickness (D374/D6988); seal strength (F88) if heat-sealable	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 4593 (Thickness of Plastic Film and Sheeting); DIN EN ISO 15106 (Water Vapour Transmission Rate of Plastic Film)	EN 1186 (Food Contact Materials); FDA 21 CFR 177 (Food Contact Plastics); EU 10/2011 (Plastic Materials Intended for Food Contact)
LLDPE Film (General Packaging)	IS 14500:1998(LINEAR LOW-DENSITY POLYETHYLENE (LLDPE) FILMS —	ISO 527-3 (Tensile properties for films and sheeting); ISO 4593 (Thickness determination by mechanical scanning); ISO 4591	ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 4593 (Thickness of Plastic Film and Sheeting); DIN EN ISO	JIS Z 1707 (Plastic Films for Packaging); GB/T 10004 (Packaging Composite Films); UL 746C (Polymeric Materials —

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	SPECIFICATION); IS 2508:2024 (Polyethylene Films and Sheets — Specification)	(Plastic Film & Sheeting – Average Thickness by Gravimetric Method);ISO 7765-1:Free-falling dart impact test for films; ISO 8295:Friction coefficient evaluation;	D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);Thickness (D374/D6988)	7765-1 (Impact Resistance by Falling Dart); DIN EN ISO 8295 (Coefficient of Friction of Plastic Films)	Outdoor Applications, where applicable)
LLDPE Stretch Film	IS 14500:1998(LINEAR LOW-DENSITY POLYETHYLENE (LLDPE) FILMS — SPECIFICATION); IS 2508:2024 (Polyethylene Films and Sheets — Specification)	ISO 4593, ISO 527-3, ISO 8295 (cling/slip behavior); ISO 7765-1 (puncture-related impact resistance).	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D5458 (Coefficient of Friction of Plastic Film and Sheeting); ASTM D4649 (Selection and Use of Stretch Wrap Films); ASTM D4321 (Package Yield of Plastic Film); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);D5748 (protrusion puncture resistance — key stretch-film spec); D5459 (load retention/elastic recovery under constant elongation); thickness	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 6383-2 (Tear Resistance of Plastic Film); DIN EN ISO 4593 (Thickness Measurement); DIN EN ISO 8295 (Coefficient of Friction of Plastic Films)	GFSI Packaging Requirements (Food Packaging Applications); JIS Z 1707 (Plastic Films for Packaging); GB/T 10457 (Stretch Film for Packaging)
HDPE Film	IS 2508:2024 (Polyethylene Films and Sheets — Specification); IS 10889:2004 (High Density Polyethylene (HDPE) Films — Specification)	ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295	Thickness; D1003 if clarity matters.; ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1894 (Standard Test	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 1183 (Density of Plastics); DIN EN ISO 4593 (Thickness Measurement); DIN EN ISO 7765-1 (Impact Resistance of Plastic Film)	JIS Z 1707 (Plastic Films for Packaging); GB/T 10004 (Plastic Composite Films); FDA 21 CFR 177 (Food Contact Applications)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor)		
Blown Polyethylene Film	IS 2508:2024 (Polyethylene Films and Sheets — Specification), IS 10889:2004 (High Density Polyethylene Films), IS 7328 (Polyethylene Material for Moulding and Extrusion — Specification)	ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295.	ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D3985 (Oxygen Gas Transmission Rate); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); Thickness; D1003	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 4593 (Thickness of Film); DIN EN ISO 6383-2 (Tear Resistance); DIN EN ISO 8295 (Coefficient of Friction)	JIS Z 1707 (Plastic Films for Packaging); GB/T 10004 (Packaging Films); FDA 21 CFR 177 (Food Contact Applications)
Cast Polyethylene Film	IS 2508:2024 (Polyethylene Films and Sheets — Specification), IS 10889:2004 (High Density Polyethylene Films), IS 7328 (Polyethylene Material for Moulding and Extrusion — Specification)	ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295.	ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); Thickness; D1003	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 4593 (Thickness Measurement); DIN EN ISO 15106 (Water Vapour Transmission); DIN EN ISO 8295 (Coefficient of Friction)	JIS Z 1707 (Plastic Films for Packaging); FDA 21 CFR 177 (Food Contact Plastics); GB/T 10004
CPP Film (Flexible Packaging)	IS 11352 (Flexible pouches for packing of vanaspati up to 2 kg/2 L); IS 12724 (Flexible pouches for packing of	ISO 17557 (Cast Polypropylene Films); ISO 18602 (Flexible Packaging);ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295.	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method);	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 6383-2 (Tear Resistance); DIN EN ISO 15106 (Water Vapour Transmission); DIN EN	JIS Z 1707 (Plastic Films for Packaging); FDA 21 CFR 177; EU 10/2011 (Food Contact Plastics)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	refined edible oils up to 5 kg/5 L); IS 14129 (Flexible packaging materials for packing vanaspati in 10 kg & 15 kg packs); IS 14636 (Flexible packaging materials for edible oils, ghee & vanaspati); IS 15609 (Polyethylene Flexible Pouches for the Packing of Natural Mineral Water and Packaged Drinking Water); IS 13244 (Rigid plastic containers for packaging liquid pesticides and agrochemicals); IS 13575 (Ethylene Vinyl Acetate (EVA) flexible films for stretch wrapping); IS 15395 (Plastics and laminates used for packaging aseptic liquid beverages); IS 10910:1984 (Polypropylene for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10951:2020 (Polypropylene (PP) Materials for Moulding and Extrusion — Specification); IS 14534:2016 (Guidelines for Recycling of Plastics); IS 7019:1982 (Glossary of Terms in Plastics and Flexible Packaging, Excluding Paper); IS 9845:1998 (Determination of Overall Migration of Plastic Materials and Articles Intended to Come into Contact with Foodstuffs — Method of Analysis); IS 9833:1981 (List of Colourants for Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS		ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting);F88 (seal strength — CPP is almost always the heat-seal layer, this is a core spec); thickness; D1003	ISO 8295 (Coefficient of Friction)	

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	2508:2024 (Polyethylene Films and Sheets — Specification); IS 13360 (Part 3/Sec 1):1995 (Determination of Thickness of Film and Sheeting by Mechanical Scanning); IS 13360 (Part 3/Sec 3):1997 (Determination of Density and Relative Density of Non-Plasticized Plastics); IS 13360 (Part 5/Sec 1):2021 (Determination of Tensile Properties — General Principles); IS 13360 (Part 5/Sec 3):1999 (Determination of Tensile Properties of Plastic Film and Sheeting); IS 13360 (Part 5/Sec 6):1999 (Determination of Impact Resistance by Free-Falling Dart Method); IS 13360 (Part 5/Sec 14):2013 (Determination of Coefficients of Friction); IS 13360 (Part 5/Sec 23):2013 (Determination of Puncture Impact Behavior of Rigid Plastics); IS 13360 (Part 6/Sec 1):2021 (Determination of Vicat Softening Temperature of Thermoplastics); IS 13360 (Part 9/Sec 1):2021 (Determination of Haze and Total Luminous Transmittance of Transparent Plastics); IS 13360 (Part 9/Sec 2):2021 (Determination of Specular Gloss); IS 13360 (Part 11/Sec 1):2013 (Determination of Wetting Tension of Plastic Film and Sheeting);				
BOPP Film (Food Packaging)	IS 16460:2016 (Biaxially Oriented Polypropylene (BOPP) Films for Packaging — Specification) [unverified	ISO 17557 (Polypropylene Films); ISO 22000 (Food Packaging);ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295;	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method);	DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 15106 (Water Vapour	FDA 21 CFR 177; EU 10/2011; JIS Z 1707

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	— could not confirm this standard exists in BIS records]; IS 11352 / IS 12724 / IS 14129 / IS 14636 (Flexible pouches & packaging materials for vanaspati, edible oils & ghee — pack sizes 2 kg/2 L, 5 kg/5 L, 10–15 kg, and general); IS 15609:2005 (Polyethylene Flexible Pouches for the Packing of Natural Mineral Water and Packaged Drinking Water); IS 10951:2020 (Polypropylene (PP) Materials for Moulding and Extrusion — Specification); IS 10910:1984 (Specification for Polypropylene and its Copolymers for its Safe Use in Contact with Foodstuffs, Pharmaceuticals and Drinking Water); IS 10171:1999 (Guide on Suitability of Plastics for Food Packaging); IS 9845:1998 / IS 9833:1981 (Food-contact safety — Overall Migration Test Method & List of Approved Colourants); IS 14534:2016 (Guidelines for Recycling of Plastics); IS 13360 — Plastics: Methods of Testing (Part 3/Sec 1:1995 Thickness by Mechanical Scanning; Part 3/Sec 3:1997 Density; Part 5/Sec 1:2021 Tensile Properties — General Principles; Part 5/Sec 3:1999 Tensile Properties of Film & Sheeting; Part 5/Sec 6:1999 Impact Resistance — Free-Falling Dart; Part 5/Sec 14:2013 Coefficient of Friction; Part 8/Sec 2:2021 Water Vapor Transmission Rate ; Part		ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting);F88 (seal strength for heat-sealable BOPP); D1003 (clarity — important for BOPP); thickness	Transmission Rate); DIN EN ISO 15105 (Gas Transmission Rate Through Plastic Films)	

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
	8/Sec 3:2021 Gas Transmission Rate ; Part 9/Sec 1:2021 Haze & Total Luminous Transmittance; Part 9/Sec 2:2021 Specular Gloss; Part 11/Sec 1:2013 Wetting Tension);				
BOPP Film (Pressure Sensitive Tape); BOPP Film (Label Film); BOPP Film (Lamination Film)	IS 13262 (Pressure-sensitive adhesive tape); IS 15495 (Self-adhesive labels/bar code labels, where applicable); IS 14636 (Flexible packaging for edible oils, ghee & vanaspati); IS 15609 (Polyethylene Flexible Pouches for the Packing of Natural Mineral Water and Packaged Drinking Water)	ISO 17557 (Polypropylene Films); ISO 29862 (Self-adhesive Tapes – Pressure Sensitive Tape); ISO 19321 (Labels); ISO 18602 (Lamination Packaging);ISO 4593 (thickness), ISO 527-3 (tensile), ISO 8295 (friction/unwind).	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D3330 (Peel Adhesion of Pressure-Sensitive Tape); ASTM D3654 (Shear Adhesion of Pressure-Sensitive Tapes); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);D2979 (rolling ball tack — standard PSA tape property alongside peel/shear you already have); D1003 for label/lamination clarity; thickness	DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 6383-2 (Tear Resistance); DIN EN ISO 8295 (Coefficient of Friction); DIN EN ISO 15105 (Gas Transmission Rate)	FINAT Test Methods (Pressure Sensitive Label Materials); PSTC Test Methods (Pressure Sensitive Adhesive Tape); JIS Z 1707
Metallized BOPP Film	IS 11352 (Flexible pouches for vanaspati); IS 12724 (Flexible pouches for refined edible oils); IS 14129 (Flexible packaging materials for vanaspati – 10 kg & 15 kg packs); IS 14636 (Flexible packaging materials for edible oils, ghee & vanaspati); IS 15609 (Polyethylene Flexible Pouches for the Packing of Natural Mineral Water and Packaged Drinking Water)	ISO 17557 (Polypropylene Films); ISO 20663 (Vacuum Metallized Flexible Packaging Materials);ISO 4593, ISO 527-3, ISO 8295	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM D3359 (Measuring Adhesion by Tape Test, for coating/metallization adhesion); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 15105 (Gas Transmission Rate); DIN EN ISO 15106 (Water Vapour Transmission); DIN EN ISO 8295 (Coefficient of Friction)	JIS Z 1707; FDA 21 CFR 177; EU 10/2011

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting)		
BOPET Film (Flexible Packaging)	IS 13181;IS 12211 (Polyethylene Terephthalate (PET) Polymers for food packaging); IS 9845:1998 (Determination of Overall Migration of Constituents of Plastics Materials and Articles Intended to Come into Contact with Foodstuffs — Method of Analysis)	ISO 4593 (Plastics — Film and Sheeting — Determination of Thickness by Mechanical Scanning), ISO 527-3 (Plastics — Determination of Tensile Properties — Part 3: Test Conditions for Films and Sheets), ISO 6383-1 (Plastics — Film and Sheeting — Determination of Tear Resistance — Part 1: Trouser Tear Method), ISO 7765-1 (Plastics Film and Sheeting — Determination of Impact Resistance by the Free-Falling Dart Method — Part 1: Staircase Methods), ISO 8295 (Plastics — Film and Sheeting — Determination of the Coefficients of Friction),	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM F88 (Seal Strength of Flexible Barrier Materials); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting);D1003 (clarity); thickness; F88 if used as seal layer	DIN EN ISO 527 (Tensile Properties); DIN EN ISO 15105 (Gas Transmission Rate); DIN EN ISO 15106 (Water Vapour Transmission); DIN EN ISO 4593 (Thickness Measurement)	FDA 21 CFR 177; EU 10/2011; JIS Z 1707
BOPET Film (Electrical Insulation)	IS 13181; IS 11298 (Part 3/Sec 3):1998 (Plastic Films for Electrical Purposes — Specifications for Individual Materials: Biaxially Oriented Polyethylene Terephthalate (PET) Films)	ISO 6721 (Electrical Insulating Plastic Films);IS 11298 (Part 3/Sec 3):1998;ISO 4593, ISO 527-3	ASTM D257 (DC Resistance or Conductance of Insulating Materials); ASTM D149 (Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);D150 (AC loss/permittivity — you included this for Capacitor Film but dropped it here; insulation-grade film needs it too);	DIN EN 60243-1 (Electrical Strength of Insulating Materials); DIN EN 60250 (Permittivity and Dissipation Factor of Electrical Insulating Materials); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 62 (Water Absorption of Plastics)	IEC 60674 (Plastic Films for Electrical Purposes); UL 746A; UL 94 (Flammability Classification)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			thickness (critical — breakdown voltage is thickness-dependent)		
BOPET Film (Capacitor Film)	IS 13181; IS 11298 (Part 3/Sec 3):1998 (Plastic Films for Electrical Purposes — Specifications for Individual Materials: Biaxially Oriented Polyethylene Terephthalate (PET) Films)	ISO 6721 (Electrical Insulating Plastic Films – Capacitor Film);IS 11298 (Part 3/Sec 3):1998; ISO 4593, ISO 527-3.	ASTM D257 (DC Resistance or Conductance of Insulating Materials); ASTM D149 (Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials); ASTM D150 (AC Loss Characteristics and Permittivity of Solid Electrical Insulation); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);Thickness (D374/D6988) — critical since dielectric strength is reported per unit thickness	DIN EN 60243-1 (Electrical Strength of Insulating Materials); DIN EN 60250 (Dielectric Properties); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 11357 (Thermal Properties of Plastics)	IEC 60384 (Fixed Capacitors Standards); UL 746A; IEC 60674
BOPET Film (Medical Packaging); Nylon (BOPA) Film (Vacuum Packaging)	IS 13181; IS 9845:1998	ISO 11607-1 (Medical Packaging); ISO 18602 (Vacuum Packaging);ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295;	ASTM F88 (Seal Strength of Flexible Barrier Materials); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting);ASTM F1929 (dye penetration seal-integrity	DIN EN 868-5 (Packaging Materials for Terminally Sterilized Medical Devices — Sealable Pouches and Reels); DIN EN ISO 11607-1 (Packaging for Terminally Sterilized Medical Devices); DIN EN ISO 527 (Tensile Properties of Plastics); DIN EN ISO 15105 (Gas Transmission Rate Through Plastic Films); DIN EN ISO 15106 (Water Vapour Transmission Rate)	FDA 21 CFR 177; EU 10/2011; USP Class VI (Medical Plastic Applications)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			test), ASTM F2096 (bubble leak test), ASTM F2338 (vacuum decay leak test).		
PVC Cling Film	IS 10151;	ISO 22000 (Food Contact Packaging); ISO 4593, ISO 527-3, ISO 8295 (cling force tied to friction)	ASTM D1593 (Nonrigid Vinyl Chloride Plastic Film and Sheeting); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D5458 (Coefficient of Friction of Plastic Film and Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);D1003 (clarity)	DIN EN 16465 (Plastics — Methods for Calibration of Thermometers Used for Testing Biodegradation, where applicable); DIN EN ISO 527 (Tensile Properties); DIN EN 1186 (Materials and Articles in Contact with Foodstuffs); DIN EN ISO 4593 (Thickness Measurement of Plastic Film)	FDA 21 CFR 177; EU 10/2011; JIS Z 1707
EVOH Barrier Film; PVDC Barrier Film	IS 14636 (Flexible Packaging for Food Products); IS 15609 (Polyethylene Flexible Pouches for the Packing of Natural Mineral Water and Packaged Drinking Water)	ISO 15105-1 (Barrier Films – Gas Transmission Properties); ISO 15105-2 (Barrier Films – Gas Transmission Properties);ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295	ASTM D3985 (Oxygen Gas Transmission Rate Through Plastic Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM F88 (Seal Strength of Flexible Barrier Materials); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting);D1003 (clarity)	DIN EN ISO 15105 (Gas Transmission Rate Through Plastic Films); DIN EN ISO 15106 (Water Vapour Transmission Rate); DIN EN ISO 527 (Tensile Properties); DIN EN 1186 (Food Contact Materials)	FDA 21 CFR 177; EU 10/2011; JIS Z 1707
Agricultural Mulch Film	IS 15351; IS 17216:2019 (Plastics — Polyethylene Mulch Films for Agricultural and Horticultural Applications — Specification);	ISO 23517 (Agricultural Plastic Films – Mulch Films);ISO 4593, ISO 527-3, ISO 7765-1	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1709 (Standard Test Methods for Impact	DIN EN 13655 (Plastics — Mulching Thermoplastic Films for Use in Agriculture and Horticulture); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 4593 (Thickness Measurement);	EN 13655 (Thermoplastic Mulching Films for Agriculture); JIS K 6781 (Agricultural Polyethylene Films)

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D5208 (Fluorescent Ultraviolet Exposure of Photodegradable Plastics, where applicable); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);D1003 (thickness)	DIN EN ISO 4892-3 (Artificial Weathering by UV Exposure)	
Greenhouse Cover Film	IS 15827;;IS 2508:2024 – general PE Films/Sheets spec	ISO 23517 (Plastics — Film for Agricultural Applications — Greenhouse Covering Films); ISO 4593, ISO 527-3, ISO 7765-1.	ASTM D1003 (Standard Test Method for Haze and Luminous Transmittance of Transparent Plastics); ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D4329 (Fluorescent UV Exposure of Plastics); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor)	DIN EN 13206 (Thermoplastic Covering Films for Use in Agriculture and Horticulture); DIN EN ISO 4892-3 (Artificial Weathering by UV Fluorescent Lamps); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 7765-1 (Impact Resistance of Plastic Film)	EN 13206 (Thermoplastic Covering Films for Use in Agriculture and Horticulture); JIS K 6781
Silage Film	IS 15827;IS 2508:2024	ISO 23517 (Plastics — Film for Agricultural Applications — Silage Stretch Films);ISO 4593, ISO 527-3, ISO 7765-1, ISO 8295	ASTM D882 (Standard Test Method for Tensile Properties of Thin Plastic Sheeting); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D4329 (UV Exposure of Plastics); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by	DIN EN 14932 (Thermoplastic Stretch Films for Wrapping Silage Bales); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 4593 (Thickness Measurement); DIN EN ISO 7765-1 (Impact Resistance of Plastic Film)	EN 14932 (Stretch Films for Wrapping Silage); JIS K 6781

Product / Application	BIS (IS)	ISO	ASTM	DIN	Other Standards
			Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); ASTM F1249 (Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor);D3985 (oxygen transmission rate)		
Geomembrane Film (HDPE); Pond Lining Film	IS 2508:2024 (Polyethylene Films and Sheets — Specification); IS 10889:2004 (High Density Polyethylene (HDPE) Films — Specification); IS 16352:2020 – HDPE Geomembrane spec (more directly on-point for this application);	ISO 10318-1 (Geosynthetics – Geomembranes); ISO 10319 (Geomembranes – Tensile Properties);IS 10889:2004 – HDPE Films Specification (direct match); ISO 4593, ISO 527-3.;IS 2508:2024 – explicitly names pond lining in scope (direct match); IS 16352:2020 also applicable if HDPE-based	ASTM D6693 (Tensile Properties of Plastic Geomembranes); ASTM D5199 (Measuring Nominal Thickness of Geosynthetic Materials); ASTM D1004 (Standard Test Method for Tear Resistance (Graves Tear) of Plastic Film and Sheeting); ASTM D5397 (Evaluation of Stress Crack Resistance of Polyolefin Geomembranes); ASTM D1709 (Standard Test Methods for Impact Resistance of Plastic Film by the Free-Falling Dart Method); ASTM D1894 (Standard Test Method for Static and Kinetic Coefficients of Friction of Plastic Film and Sheeting); ASTM D1922 (Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method); ASTM D2103 (Standard Specification for Polyethylene Film and Sheeting); D4833 (index puncture resistance — a standard geomembrane qualification test, missing here)	DIN EN 13361 (Geosynthetic Barriers — Characteristics Required for Use in Reservoirs and Dams); DIN EN 13362 (Geosynthetic Barriers for Canals); DIN EN ISO 527 (Tensile Properties); DIN EN ISO 12236 (Static Puncture Resistance of Geosynthetics)	GRI GM13 (Geosynthetic Research Institute — HDPE Geomembranes); GRI GM17 (LLDPE Geomembranes)

Source field: Plastic and Plastic Products